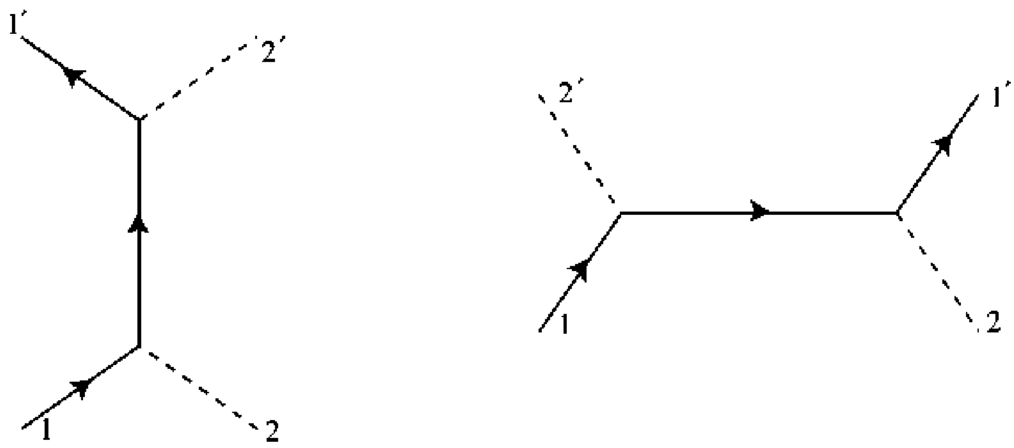

Figure 6.6. The connected second-order diagrams for boson–fermion scattering in a theory with interaction (6.1.18). Straight lines are fermions; dashed lines are neutral bosons.

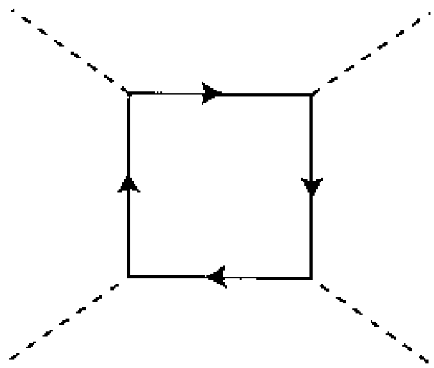

Figure 6.7. The lowest-order connected diagram for boson–boson scattering in a theory with interaction (6.1.18). Such fermion loop graphs yield an extra minus sign, arising from permutations of the paired fermion fields.

Theory I

Consider the theory of fermions and self-charge-conjugate bosons with interaction (6.1.18). The lowest-order connected diagrams for fermion–boson scattering are shown in Figure 6.6. Following the rules outlined in Figure 6.1, the corresponding S-matrix element is

$$
\begin{aligned}
&S_{\mathbf{p}_1'\sigma_1'n_1'\,\mathbf{p}_2'\sigma_2'n_2',\;\mathbf{p}_1\sigma_1n_1\,\mathbf{p}_2\sigma_2n_2} = \\
&(2\pi)^{-6}\sum_{k'l'm'klm}(-i)^2 g_{l'm'k'}\,g_{mlk}\,u^*_{l'}(\mathbf{p}_1'\sigma_1'n_1')\,u_l(\mathbf{p}_1\sigma_1n_1)\\
&\times\int d^4x\int d^4y\,\Big(-i\Delta_{m'm}(y-x)\Big)e^{-ip_1'\cdot y}e^{ip_1\cdot x}\\
&\times\Big[e^{-ip_2'\cdot y}u^*_{k'}(\mathbf{p}_2'\sigma_2'n_2')e^{ip_2\cdot x}u_k(\mathbf{p}_2\sigma_2n_2)\\
&+\;e^{-ip_2'\cdot x}u^*_{k}(\mathbf{p}_2'\sigma_2'n_2')e^{ip_2\cdot y}u_{k'}(\mathbf{p}_2\sigma_2n_2)\Big]\,. \qquad (6.1.27)
\end{aligned}
$$